II. SPACE ACT AGREEMENTS GENERALLY

The Act provides NASA with the ability to

> enter into and perform such contracts, leases, cooperative agreements, or *other transactions* as may be necessary in the conduct of its work and on such terms as it may deem appropriate, with any agency or instrumentality of the United States, or with any State, Territory, or possession, or with any political subdivision thereof, or with any person, firm, association, corporation, or educational institution.[7]

This key phrase provides NASA with tremendous flexibility to accomplish its goals. Countless Federal government agencies have the ability to enter into contracts, grants, and cooperative agreements, but not all enjoy the OTA.[8]

The Federal Grant and Cooperative Agreement Act[9] specifies when the Federal Government should use each particular type of legal instrument. In general, when the principal purpose of the instrument is to acquire property or services for the direct benefit or use of the United States Government, agencies must use a procurement contract.[10] Executive Branch agencies use grants when the principal purpose of the relationship is to transfer a thing of value to carry out a public purpose of support or stimulation and substantial involvement with the partner is not expected.[11] When the Government does so and substantial involvement is expected, it uses a cooperative agreement.[12]

The Federal Acquisition Regulation (FAR) provides a well-understood framework for conducting a Federal procurement. Those rules are supplemented by the NASA Federal Acquisition

---

[7] 42 U.S.C.A. § 2473(c)(5) (emphasis added).

[8] GAO has recognized that, since 1958, Congress has extended the authority to use "other transactions" to a number of other Federal agencies, including the Department of Defense [10 U.S.C.A.§ 2371 (1998)], the Department of Homeland Security [6 U.S.C.A. § 391 (2007 and Supp. 2008], the Department of Energy [42 U.S.C.A. § 7256(g)(1) (Supp. 2008)], and the Department of Transportation [49 U.S.C.A. § 106(l) (2007)]. Matter of Exploration Partners, LLC, B-298804, 2006 CPD 201, at 4, n.1 (2006) [hereinafter *Partners*], http://www.gao.gov/decisions/bidpro/298804.pdf.

[9] 31 U.S.C.A. §§ 6301-6308 (2003).

[10] 31 U.S.C.A. § 6303 (2003).

[11] 31 U.S.C.A. § 6304 (2003).

[12] 31 U.S.C.A. § 6305 (2003).

Regulation Supplement. In combination, these regulations impose a wide range of requirements on the Agency when it acquires goods or services, including the need for advertising, competition, source selection, implementation of socio-economic goals, and contract administration. The rules also provide a well understood framework for challenging NASA contract awards both before the Agency and GAO. While many of our clients might like it to be otherwise, acquisition efficiency is not one of the chief characteristics of this regulatory environment.

By contrast, Space Act agreements (hereinafter SAAs), are not procurements and are not subject to this framework. As such, when the factual environment supports the use of such agreements, they can be developed and executed quickly, on both an exclusive and non-exclusive basis, in response to an Agency request or an unsolicited proposal, and on a reimbursable or non-reimbursable basis. They are not, however, a substitute for procurements conducted under the FAR when the Agency is acquiring goods or services for its direct benefit. As attorneys, we are sometimes called upon to convey this message to clients, or outside entities, who believe the Act can be used to avoid a statutory and regulatory framework, or to accomplish an acquisition objective more quickly.

SAAs are most suited to the accomplishment of mutually agreeable goals within the Agency's mission. Examples include partnerships with other Federal agencies, State and local governments, educational institutions, for profit companies, and not-for-profit entities. One of the most common uses is provision of NASA facilities[13] on a non-interference basis to allow technological or scientific development, such as use of Agency wind tunnels at Langley Research Center and Ames Research Center, or use of thermal-vacuum test chambers at Goddard Space Flight Center, or Johnson Space Center. NASA facilities are often unique, due to size, capability, or location, such that opportunities for similar work outside the Agency do not exist. Hence, outside entities often need and seek out such capability from NASA. Indeed, one of the few constraints on development

[13] Another common use is provision of NASA personnel with unique knowledge, capabilities, and skills.

of SAAs for use of NASA facilities is the prohibition on Government competition with the private sector in provision of facilities or equipment.[14]

III. SPACE ACT AGREEMENT POLICY AND EXECUTION

In years past, because there was no established procedure for creating, negotiating, and executing SAAs, the process was often time and resource intensive. Today, the Agency employs an efficient computer based program called Space Act Agreement Maker (SAAM).[15] Following entry of correct factual information into the system by the client, assisted by an agreements manager, SAAM automatically generates the first draft of an agreement containing appropriate provisions. This ensures that all statutory and regulatory requirements are included and, analogous to the use of FAR clauses in a procurement contract, provides a measure of reassurance to reviewing Agency attorneys. Drafts generated by SAAM may then be individually tailored to include any situation specific facts and provisions. Provided both NASA and its outside entity have a good understanding of the arrangement, participate in good faith in early and comprehensive discussions both before and after generation of the first draft, the SAAM process can lead to formation and signature of final agreements within weeks. This is a great benefit to both sides. SAAM also allows for the archiving, search, and retrieval of existing agreements.

NASA Policy Directive (NPD) 1050.1H provides the regulatory framework for reimbursable, non-reimbursable, and funded SAAs.[16] The NPD defines each type of agreement, indicates which issues must be addressed in every agreement, and specifies which individuals have authority for signing agreements on

---

[14] The Commercial Space Competitiveness Act indicates the Government may allow non-Federal entities to use space-related facilities on a reimbursable basis if, amongst other things, "equivalent commercial services are not available on reasonable terms." 15 U.S.C.A. § 5807 (1998).

[15] SAAM is currently accessible by registered internal NASA users.

[16] *See* NASA Policy Directive, *Subject: Authority to Enter into Space Act Agreements*, NPD 1050.1H (Nov. 29, 2006), *available at* http://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPD&c=1050&s=1H.

behalf of the Agency.[17] Reimbursable SAAs are those wherein NASA's costs associated with the undertaking are reimbursed by the Agreement partner in whole or in part.[18] Nonreimbursable SAAs involve NASA and one or more partners in a mutually beneficial activity that furthers an Agency mission, wherein each party bears the cost of its participation and there is no exchange of funds between the parties.[19] Funded SAAs are those under which NASA transfers appropriated funds to a domestic partner to accomplish an Agency mission.[20]

Some of the more important areas that must be covered in an agreement include (1) respective responsibilities of the parties, (2) responsibilities or performance mile-stones, (3) clearly defined financial commitments (if any), including a statement that NASA's performance of the Agreement is subject to the availability of appropriated funds and that no provision of the Agreement shall be interpreted to require obligation or payment of funds in violation of the Anti-Deficiency Act,[21] (4) resource commitments, providing that NASA usage of facilities, equipment, and personnel shall have priority, (5) allocation of risk between the parties, (6) allocation of intellectual property rights, (7) termination rights and obligations, and (8) a fixed expiration date.[22]

Based on the wealth of previously executed SAAs, and experience interpreting issues arising under various provisions contained in those agreements, NASA has also prepared the "Space Act Agreements Guide," (the Guide) administered by the Office of General Counsel at NASA Headquarters.[23] The Guide is intended to explain NASA agreement practice and provide assistance to those involved in formation and execution of SAAs.

---

[17] *Id.*
[18] *Id.*
[19] *Id.*
[20] *Id.*
[21] 31 U.S.C.A. § 1341 (2003).
[22] *Authority to Enter into Space Act Agreements*, *supra* note 16.
[23] *See* NASA Advisory Implementing Instruction, *Space Act Agreements Guide*, NAII 1050-1A, effective August 15, 2008, *available at* http://nodis3.gsfc.nasa.gov /NPD_attachments/NAII_1050_1A.doc.

It does not, in and of itself, establish substantive or procedural requirements contained in other authorities.

IV. EXAMPLES

NASA has used its OTA for a wide variety of innovative and useful purposes. Beginning in 2001, NASA entered into a number of non-reimbursable SAAs with Bigelow Aerospace (hereinafter BA) for collaboration, technical and short-term personnel exchanges, as well as licensing agreements involving payment of royalties. BA is developing a line of inflatable orbital space habitats based on NASA patented technology which, if successful, will help fulfill NASA's charter "to encourage to the maximum extent possible the fullest commercial use of space,"[24] and help the Agency to achieve its own long-term goals of establishing a human presence on the Moon and Mars. In addition to its original financial investment in facilities and personnel, BA has paid for the launch of two sub-scale demonstrators. JSC engineers who participate in interactions with BA gain insight into state of the art capabilities concerning the development, launch, control, and operations of expandable space structures. The arrangements promise to be very much a win-win situation for both sides.

Other examples include routine SAAs at Langley Research Center for support to the aerospace industry, such as blended wing body and fighter aircraft research, as well as air traffic control. More unusual agreements support (1) collaboration with Jamestown 2007, planners of the commemoration of the 400th anniversary of the landing at Jamestown, Virginia, to develop educational resources, exhibits, and support materials focusing on themes of exploration past, present, and future, (2) reimbursable wind-tunnel work testing the surface friction of more than 60 types of fabric to help develop the "world's fastest swimsuit" used in the summer Olympic games in Beijing,[25] and

---

[24] *See* 42 U.S.C.A. § 2473(a)(3) (2003).

[25] In August 2008, swimmer Michael Phelps used Speedo's LZR Racer while winning eight Olympic gold medals and setting seven world records. For more information *see*, NASA News & Features, *Olympic Swimmers Shattering Records in NASA-Tested*

(3) reimbursable work for the filming of major motion pictures, that has occurred at several NASA centers.

Besides the OTA under the Act, NASA has also taken advantage of its newly provided authority under the enhanced use lease demonstration program.[26] That legislation allowed the Agency to lease out underutilized real property at fair market value and retain proceeds obtained from the lessee. Ames Research Center and Kennedy Space Center have entered into a variety of enhanced use leases bringing to their respective campuses other Governmental agencies, universities, and private parties. As one example, Ames entered into an enhanced use lease with the Air Force of the National Full-Scale Aerodynamics Complex (NFAC), a facility consisting of the 40 x 80 foot and 80 x 120 foot wind tunnels. The NFAC was underutilized by NASA and there was insufficient funding to preserve, operate, and upgrade the facility. The lease with the Air Force allowed cost recovery authorized by the statute, which preserved this national asset for the benefit of the country. Ames currently has leases with 14 universities and approximately 40 private companies.[27]

V. FUNDED SPACE ACT AGREEMENTS

In certain circumstances, SAAs can be used to provide funding to an outside partner. NASA recently developed funded SAAs with for-profit companies to implement Phase One of the Commercial Orbital Transportation Services (hereinafter, COTS) project. In accordance with United States Space Exploration Policy, NASA must complete assembly of the *Interna-*

Suit (Aug. 13, 2008), http://www.nasa.gov/topics/technology/features/2008-0813-swimsuit.html.

[26] 42 U.S.C.A. § 2459j (2003 and Supp. 2008).

[27] Enhanced use leases can be a significant means of leveraging the value of an agency's real property and facilities. As NASA struggles with its portfolio of aging infrastructure, coupled with insufficient resources for construction of new facilities, such leveraging could be important. In addition to the OTA and enhanced use leasing authority under the Space Act, the Agency may also use the National Historic Preservation Act (NHPA) for this purpose. 16 U.S.C. §§ 461-470x-6 (2000 and Supp. 2008). Proceeds of leases under the NHPA may also be retained by the Agency and used to "defray the costs of admini-stration, maintenance, repair, and related expenses" of properties on the National Register of Historic Places. 16 U.S.C. § 470h-3(b) (2000).

tional Space Station (*ISS*), planned for the end of the decade, using the *Space Shuttle* and then retire the *Shuttle.*[28] Once the *Space Shuttle* is retired, the Russian *Soyuz* vehicle will be the only one available for crew exchange and rescue services on the *ISS* until the *Orion* spacecraft or a United States commercially provided crew service is available.[29] United States space policy directs the purchase of commercially available U.S. space transportation products and services to the maximum extent possible, consistent with mission requirements and applicable law.[30] NASA's policy is to employ U.S. commercial services for both cargo and crew exchange at the earliest available opportunity, while minimizing the technical risks of interrupting the U.S. crew presence on orbit, or leaving the *ISS* in a state of disrepair because failed components cannot be replaced.[31] To this end, NASA is investing $500 million to stimulate the commercial space industry and to facilitate U.S. industry demonstration of commercial space transportation capabilities under Phase One of the COTS project.[32] Phase Two of the COTS project is a planned competitive procurement of orbital transportation services to resupply the *ISS* with crew and cargo.

NASA's application of the OTA in support of the COTS project was fully explored recently before GAO. While many had always believed that one of the most useful characteristics of SAAs using the OTA is that such agreements are not subject to protest since they are not procurements, that view was tested and reinforced in two GAO decisions examining the COTS project.[33]

---

[28] *See* United States Space Transportation Policy, Fact Sheet (Jan. 6, 2005), *available at* http://www.ostp.gov/galleries/Issues/Space_Transportation_Policy05.pdf.

[29] *See* Statement of Michael D. Griffin, Administrator, National Aeronautics and Space Administration, before the Subcommittee on Space, Aeronautics, and Related Sciences, of the Commerce, Science, and Transportation Committee of the United States Senate, (Nov. 15, 2007) [(hereinafter Statement], *available at* http://commerce.senate.gov/public/_files/CopyofNASAISSTransportationTestimonyRevd FINAL.pdf.

[30] *See* United States Space Transportation Policy, *supra* note 28.

[31] *See* Statement, *supra*, note 29.

[32] *Id.*

[33] *Partners*, *supra* note 8; Matter of Rocketplane Kistler, B-310741, 2008 CPD ¶22 (2008) [hereinafter *Rocketplane*].

In the first case, Exploration Partners challenged NASA's decision to provide funded SAAs totaling $500 million over a four-year period to Space Exploration Technologies (SpaceX) and Rocketplane Kistler under the COTS project.[34] GAO recounted NASA's decision to carryout the project in two phases, the first being a "period of development and demonstration by private industry, in coordination with NASA, of various space transportation capabilities . . . determined to be most desirable for the Government and other customers."[35] Phase Two was described as a "potential competitive procurement of orbital transportation services to resupply the [*ISS*] with cargo and crew, if a capability is successfully demonstrated and the Government determines it is in its best interests."[36] Each agreement was characterized by performance milestones proposed by the participants and negotiated with NASA.[37] NASA's contribution was a fixed amount regardless of each participant's ability to raise additional private funding.[38] Exploration Partners claimed it was "the only company that offered a fully funded end-to-end transportation system" and should have received an agreement or, in the alternative, that the "COTS program[39] . . . be re-bid under the original terms and conditions without interference in obtaining Shuttle hardware, cost data or interference in commercial business relationships."[40]

GAO concluded that under the Competition in Contracting Act of 1984 and its own bid protest regulations, it would not review issuance of SAAs pursuant to NASA's OTA because the issuance of such agreements was not tantamount to the award of contracts for the procurement of goods and services and was, therefore, outside GAO's bid protest jurisdiction.[41] GAO noted that the Act did not define other transactions but that Congress

---

[34] *Partners, LLC, supra* note 8, at 2.

[35] *Id.*

[36] *Id.*

[37] *Id.*

[38] *Id.*

[39] NASA is carrying out the COTS *project* under the Commercial Crew and Cargo *Program*. *See* NASA, *Commercial Crew & Cargo, available at* http://www.nasa.gov/offices/c3po/home/index.html (last visited Nov. 13, 2008).

[40] *Partners, supra* note 8, at 3.

[41] *Id.* at 4.

had recognized at the time of its promulgation that this was a grant of "'broad authority.'"[42] In reaching this conclusion, GAO applied the rule of statutory construction that indicates a statute ought to be so construed that, if it can be prevented, no clause, sentence, or word shall be superfluous, void, or insignificant.[43] GAO noted that Exploration Partners had not timely challenged NASA's original decision to award SAAs since such a challenge was required to be filed before the closing date set for receipt of proposals.[44]

In the second case, Rocketplane Kistler challenged NASA's decision to continue Phase One development of the COTS program with a second round of funded SAAs with industry.[45] NASA announced it was soliciting proposals for demonstrations involving an end-to-end transportation system of services including ground operations and integration, launch, rendezvous, proximity operations, docking or berthing, orbital operations reentry, and safe disposal or return.[46] Participants were informed they were expected to secure funding necessary to complete the proposed capability demonstration, but funding could include up to $174 million from NASA.[47] Payments would be made on completion of performance milestones.[48] Rocketplane argued that the principal purpose of the announcement is to obtain research and development services for the direct benefit of NASA and that NASA was required to use a procurement contract.[49]

NASA argued that, through the agreements, it obtained no vehicles, supply service, prototype, hardware, or other property, no systems or vehicle designs, and only the minimum Government-purpose data rights legally required by the Act.[50] The announcement did not provide NASA any right to future use of

---

[42] *Id.*
[43] *Id.*
[44] *Id.* at 6.
[45] *Rocketplane*, *supra* note 33, at 1-2.
[46] *Id.* at 2.
[47] *Id.* at 2-3.
[48] *Id.* at 3.
[49] *Id.*
[50] *Id.* at 4.

systems and vehicles developed and demonstrated.[51] The participant proposed the capabilities it would demonstrate and established the technical and schedule milestones for the demonstrations.[52] GAO noted that the announcement here sought to "incentivize the private sector to develop and demonstrate their own commercial technologies."[53] GAO concluded the announcement did not principally provide for the acquisition of goods and services for the direct benefit and use of NASA.[54] Rather the record supported the agency's arguments that the principal purpose of the announcement was to encourage, support and stimulate the development of a commercial market for space transportation, from which NASA could potentially acquire orbital transportation services.[55]

VI. CONCLUSION

Practical and strategic benefits aside, perhaps the most important aspects of the Act are intangible. NASA's ability to partner with outside entities can facilitate the co-location and interaction of intellectual talent. Close physical collaboration provides the potential for the promotion and development of unexpected scientific and technological innovations, important goals for an Agency charged with opening up the solar system. Also less obvious is the creation of a vibrant and creative work environment conducive to the attraction of a talented workforce, in particular for employees contemplating new careers. The best work force is often attracted to state-of-the-art facilities and exciting projects, not necessarily achievable through other means.

When the chief architect of the Act was once asked how the phrase, "other transactions," came about, he replied,

> Well, I tried to cover everything else that was [raised by others]. When somebody said, well, suppose we have this kind of

---

[51] *Id.*
[52] *Id.*
[53] *Id.* at 4-5.
[54] *Id.* at 5.
[55] *Id.* NASA subsequently entered into a funded SAA with Orbital Sciences Corporation to conduct Phase One of the COTS project.

> a transaction or that kind of a transaction, I figured, it may not be covered under contracts, leases, and cooperative agreements. I couldn't think of any other terminology to use, so I used "other transactions as may be determined or necessary in the conduct of its work." So it was a sort of catchall phrase that I tried to use. . . . [A]n "other transaction" is not a procurement contract, cooperative agreement, or grant and, therefore, is *not subject to the laws, regulations, and other requirements applicable to such contracts, agreements, and grants. It is this flexibility which provides authority to structure agreements in accordance with standard business practices.*[56]

NASA's current use of that implementing language, drafted more than fifty years ago, and reinforced by recent GAO decisions, constitutes a lasting legacy to that legislative prowess.

---

[56] Telephone Interview with Paul G. Dembling (July 25, 2002) (transcript on file with author) (emphasis added).

REFLECTIONS UPON THE NOTION OF LIABILITY: THE INSTANCES OF *KOSMOS 954* AND SPACE DEBRIS

Andrew Brearley[*]

INTRODUCTION

This piece seeks to examine the principles of space law and how they could potentially be applied to the present problem of space debris.[1] In order to do so it conducts an inquiry into the events surrounding the crash of a nuclear powered satellite into Canadian territory. The incident has now been largely forgotten, although occasional references to it appear in popular Canadian culture,[2] but it remains the most useful instance in which the liability provisions of space law have been tested.

The United Nations Office of Outer Space Affairs (OOSA) highlights two specific security issues which affect humanity's activities in space: the use of nuclear power and space debris. Although it is debris which currently raises the greatest concerns, the use of nuclear power has resulted in the testing of the notion of liability as it is contained within space law.

The two issues present different legal and technical challenges, principally due to the locations in which they pose a danger. Debris may be a threat on the ground should it survive re-entry to the atmosphere, however this threat is negligible in

---

[*] Andrew Brearley recently completed his Ph.D. examining the legal and policy implications of orbital debris at the University of Southhampton (U.K.). He has previously published on the subjects of debris and the ownership of extra-terrestrial resources.

[1] Space debris, or orbital debris, is defined as artificial objects trapped in Earth orbit. Due to the extreme speed at which they travel they are considered to pose a serious threat to active space projects. *See* NICHOLAS L. JOHNSON & DARREN S. MCKNIGHT; ARTIFICIAL SPACE DEBRIS (Krieger Publishing Company, 1991); *Technical Report on Space Debris* (New York: United Nations, 1999), www.unoosa.org/pdf/reports /ac105/AC105_720E.pdf.

[2] *See e.g.*, DOUGLAS COUPLAND, ELEANOR RIGBY 192 (London, Harper Perennial, 2005) (a character discovers a piece of radioactive debris in Canada from a Soviet era nuclear powered satellite).

comparison to the risk which is poses in orbit. In contrast nuclear power sources pose the greatest concern on ground level. Legally the two problems have to be considered separately as a differing degree of liability exists depending upon where an incident occurs. A further differentiation between the two problems concerns remedial action, addressing one problem can have a detrimental effect with reference to the other.

In a typical year more than fifty large objects fall back to the Earth from orbit.[3] Simply due to the geography of the Earth the majority of objects, both natural and artificial, that enter the atmosphere, and survive the process of re-entry, land in the oceans.[4] Larger objects, such at the Soviet *MIR* space station, re-enter the atmosphere in a controlled fashion, and what remains of the object after "burning-up" is deliberately targeted at the high seas. However, such instances demonstrate that remedial action towards the debris problem in orbit, specifically bringing large objects back into the atmosphere, can result in physical danger being caused on the surface of the planet.

Almost five decades of space travel have produced few instances of damage being caused by objects returning from space. However, the presence of debris in Earth orbit creates the increased possibility of the Liability Convention[5] being used as a means to resolve international disputes. Despite having been in effect for several decades, the treaty is largely untested. However, it was exercised, although not to its full extent, in a case between the USSR and Canada.

BACKGROUND TO THE CLAIM

On January 24, 1978 a Soviet satellite, *Kosmos 954*,[6] accidentally crashed into north western Canada. The fragmented parts of the satellite were scattered over an area the size of Aus-

---

[3] M. Yakovlev; *The "IADC Space Debris Mitigation Guidelines" and Supporting Documents*, *in* PROCEEDINGS OF THE 4TH EUROPEAN CONFERENCE ON SPACE DEBRIS 595 (Darmstadt, Germany: European Space Agency, 2005).

[4] S. Neil Hosenball, *Nuclear Power Sources In Outer Space,* 6(2) J. OF SPACE L. 119 (1978).

[5] Convention on International Liability for Damage Caused by Space Objects, Mar. 29, 1972, 28 U.S.T. 2389, T.I.A.S. No. 7762 [hereinafter Liability Convention].

[6] The satellite is also often referred to as "Cosmos 954".

tria,[7] and included some of its power source, 30 kilograms of enriched uranium.[8] These events, and the subsequent claim for damages, made by the government of Canada, provide an insight into the notion of liability as it is contained within space law, revealing difficulties inherent in its application.

Kosmos 954 was launched on September 18, 1977.[9] The Convention on Registration of Objects Launched into Outer Space,[10] requires that every object "launched into Earth orbit or beyond" is registered with the Secretary-General of the United Nations, and that a registry shall be maintained of all such objects,[11] this is to include the "[g]eneral function of the space object."[12] The purpose of the treaty is to create a mandatory international database which would "assist in their identification and would contribute to the application and development of in-

---

[7] U.S. Congress, Office of Technological Assessment, *Orbiting Debris, A Space Environmental Problem - Background Paper*, 3 (Washington D.C.: U.S. Government Printing Office, 1990).

[8] Nicholas L. Johnson, *Monitoring and Controlling Debris in Space*, SCIENTIFIC AMERICAN, 64 (Aug. 1998).

[9] *See* Committee on the Peaceful Uses of Outer Space, Information Furnished in Conformity with General Assembly Resolution 1721 B (XVI) by States Launching Objects into Orbit or Beyond, U.N. Doc. A/AC.105/INF.368 (Nov. 22, 1977) [hereinafter Information Furnished in Conformity with GA Res 1721 B (XVI)], http://www.unoosa.org/pdf/inf368E.pdf; Committee on the Peaceful Uses of Outer Space, Note Verbale Dated 3 March 1978 From the Permanent Representative of Canada Addressed to the Secretary-General, U.N. Doc. A/AC.105/217 (Mar. 6, 1978) [hereinafter Note Verbale Dated 3 March 1978], http://www.unoosa.org/pdf/reports/ac105/AC105_217E.pdf; and Committee on the Peaceful Uses of Outer Space, Note Verbale Dated 19 December 1978 from the Permanent Representative of Canada to the United Nations Addressed to the Secretary-General, U.N. Doc. A/AC.105/236 (Dec. 22, 1978) [hereinafter Note Verbale Dated 19 December 1978], http://www.unoosa.org/pdf/reports/ac105/AC105_236E.pdf.

This document was submitted to the United Nations in accordance with the Convention on Registration of Objects Launched into Outer Space, which commits states to register objects which are launched into space with the Secretary General of the United Nations. *See* Convention on Registration of Objects Launched into Outer Space, Sep. 15, 1976, 28 U.S.T. 695, 1023 U.N.T.S. 15 [hereinafter Registration Convention]. An initial press report of the launch, before the satellite crashed, quoted the launch date as 19th September. *See*, *Soviets Launch New Salyut*, AVIATION WEEK & SPACE TECHNOLOGY (Oct. 3, 1977).

[10] Registration Convention, *supra* note 9.

[11] *Id.* at art. II, para. 1.

[12] *Id.* at art. IV, para. 1.